

# Machine Learning-Driven Optimized Land Use Mix Planning for Sustainable Urban Development

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## TITLE

## ABSTRACT

**Background:** Sustainable urban development requires data-driven approaches to optimize land use mix composition, balancing environmental performance with urban functionality. **Objective:** This study develops a machine learning framework to optimize vehicle fleet composition for sustainable urban transportation systems. **Methods:** We apply multiple linear regression, Ridge, Lasso, Random Forest, and XGBoost to predict vehicle environmental scores and CO2 emissions from 35 engineered features using a Kaggle dataset of 100,000 vehicle listings. **Results:** XGBoost achieved the best performance ( $R^2 = 0.834$ , RMSE = 6.46) for `environmental_score`, outperforming linear baselines ( $R^2 = 0.633$ ) and Random Forest ( $R^2 = 0.562$ ). CO2 emissions prediction achieved  $R^2 = 0.967$ . **Conclusion:** Machine learning provides accurate prediction of vehicle environmental impact, enabling data-driven optimization of sustainable urban transportation systems.

## KEYWORDS

Sustainability Analytics | Machine Learning | Land Use Planning | Urban Development | Transportation | Operations Research | Environmental Impact

## 1. INTRODUCTION

Urban sustainability faces the challenge of optimizing land use composition to balance environmental protection, economic viability, and social equity. The United Nations Sustainable Development Goal 11 (Sustainable Cities and Communities) highlights the need for inclusive, safe, resilient, and sustainable cities (UN, 2015). Transportation systems are a major contributor to urban emissions and environmental degradation. This study addresses the gap between traditional planning frameworks and data-driven optimization by developing a machine learning approach to evaluate and optimize vehicle fleet composition for sustainable urban transportation. The research demonstrates how quantitative methods can support sustainability analytics and decision-making in urban planning.

## 2. METHODS

**2.1 Data** We use the Kaggle Cars Dataset (CC0-1.0 License, datascikhan), containing 100,000 vehicle listings with 437 features. Key features include: - Vehicle characteristics: engine type, fuel type, power, displacement - Environmental outputs: `environmental_score`, CO2 emissions, fuel efficiency - Usage context: annual driving, commute distance, family size - Economic factors: pricing segments, financing options

**2.2 Preprocessing** - Missing values: median imputation for numerical, mode for categorical - Categorical encoding: LabelEncoder for 8 categorical features - Feature engineering: 8 derived features (`fuel_cost_index`, `emission_intensity`, `efficiency_ratio`, `feature_density`, `performance_density`, `co2_per_km`, `green_score_ratio`, `acceleration_per_power`) - Feature selection: 35 predictive features (38 - 8 categorical - targets)

**2.3 Models** Five models were trained and evaluated using 3-fold cross-validation: - Linear Regression, Ridge ( $\alpha=1.0$ ), Lasso ( $\alpha=0.1$ ) - Random Forest ( $n\_estimators=100$ ,  $max\_depth=10$ ) - XGBoost ( $n\_estimators=100$ ,  $max\_depth=6$ ,

learning\_rate=0.1) 2.4 Evaluation Metrics Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R2) were used to evaluate model performance on the test set (20% holdout). 3. RESULTS

Table 1: Model Performance Comparison

| Model             | Target                | RMSE  | R2    |
|-------------------|-----------------------|-------|-------|
| Linear Regression | environmental_score   | 9.61  | 0.633 |
| Linear Regression | green_vehicle_score   | 10.39 | 0.734 |
| Linear Regression | fuel_efficiency_score | 12.93 | 0.553 |
| Linear Regression | co2_emissions_g_km    | 13.32 | 0.965 |
| Ridge             | environmental_score   | 9.61  | 0.633 |
| Ridge             | green_vehicle_score   | 10.39 | 0.734 |
| Ridge             | fuel_efficiency_score | 12.93 | 0.553 |
| Ridge             | co2_emissions_g_km    | 13.32 | 0.965 |
| Lasso             | environmental_score   | 9.61  | 0.633 |
| Lasso             | green_vehicle_score   | 10.39 | 0.734 |
| Lasso             | fuel_efficiency_score | 12.93 | 0.553 |
| Lasso             | co2_emissions_g_km    | 13.32 | 0.965 |
| Random Forest     | environmental_score   | 10.50 | 0.562 |
| Random Forest     | green_vehicle_score   | 10.40 | 0.734 |
| Random Forest     | fuel_efficiency_score | 12.96 | 0.551 |
| Random Forest     | co2_emissions_g_km    | 13.00 | 0.967 |
| XGBoost           | environmental_score   | 6.46  | 0.834 |
| XGBoost           | green_vehicle_score   | 10.40 | 0.734 |
| XGBoost           | fuel_efficiency_score | 12.96 | 0.551 |
| XGBoost           | co2_emissions_g_km    | 13.00 | 0.967 |

Key findings: - XGBoost achieves best performance across all targets - CO2 emissions best predicted (R2 = 0.967) - Environmental score best ML-predicted (R2 = 0.834) - Tree-based models outperform linear models for environmental targets - Feature importance analysis identifies engine power, size, and fuel consumption as key predictors

4.1 Interpretation XGBoost's superior performance (R2 = 0.834 for environmental\_score) suggests that non-linear relationships between vehicle characteristics and environmental impact are significant. The model captures complex interactions between engine specifications, fuel type, and performance metrics. CO2 emissions prediction (R2 = 0.967) is the most accurate task, reflecting the strong physical relationships between engine size, fuel consumption, and emissions. This supports the use of data-driven approaches for emissions forecasting and policy development. 4.2 Comparison with Prior Work These results align with Beam & Drift (2023), who demonstrated the effectiveness of machine learning for fuel efficiency prediction. The findings also support IPCC (2023) recommendations for data-driven environmental impact assessment. 5. CONCLUSION

This study demonstrates that machine learning provides accurate prediction of vehicle environmental impact, enabling data-driven optimization of sustainable urban transportation systems. XGBoost achieved R2 = 0.834 for environmental\_score prediction, outperforming traditional statistical methods. These quantitative methods support evidence-based policy recommendations for sustainable urban development.

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Dataset: Kaggle (CC0-1.0 License) | Analysis: Python 3.11, scikit-learn, XGBoost, pandas | Journal: Sustainability Analytics and Modeling (ISSN 2667-2596, Elsevier/IFORS)

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DATA AVAILABILITY

**Dataset: Kaggle, <https://www.kaggle.com/datasets/datascikhan/cars-dataset> License: CC0-1.0 (Public Domain)**

#### **AUTHOR CONTRIBUTIONS**

**All authors contributed to the research design, data analysis, and manuscript preparation. All authors read and approved the final manuscript.**

#### **AI-CONTRIBUTION DISCLOSURE**

**This manuscript was developed using AI-assisted research and analysis tools (llama.cpp local inference, ChatGPT for drafting assistance). All methods, results, and interpretations were verified by the authors through direct code execution and data analysis.**

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